Datasets

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The datasets used in the course exercises are organized by week. Click on the filenames to download the data files.

1 Week 45

Dataset	Exercise	Description
averag-prope-amino-acids.xlsx	Average properties of amino acids and proteins.	Amino acid frequency data
uv-spec-apo-holo-myo.csv	UV spectrum of apo- and holo-myoglobin	UV spectra for apo and holo myoglobin
trypt-absor-fluor-emission.xlsx	Tryptophan absorption and fluorescence	Tryptophan fluorescence emission data
trypt-absor-fluor-extinction.xlsx	Tryptophan absorption and fluorescence	Tryptophan fluorescence extinction data
trypt-absor-fluor-ph2.xlsx	Tryptophan absorption and fluorescence	Tryptophan fluorescence at pH 2
trypt-absor-fluor-ph7.xlsx	Tryptophan absorption and fluorescence	Tryptophan fluorescence at pH 7
extin-coeff-human-myogl.txt	Extinction coefficient of human myoglobin	Titin protein sequence data
the-fluor-protei-mcherr.csv	The fluorescent protein “mCherry”.	mCherry fluorescent protein spectra
protei-blood-plasma.xlsx	Proteins in blood plasma.	Protein concentration in blood plasma data

2 Week 46

Dataset	Exercise	Description
dialys-exper.xlsx	Dialysis experiment	Dialysis experiment data
adp-bindin-pyruva-kinase.csv	ADP binding to pyruvate kinase.	ADP binding to pyruvate kinase data

Dataset	Exercise	Description
inter-bindin-data.xlsx	Interpretation of binding data.	Binding data for interpretation
deter-type-streng-coope.xlsx	Determination of type and strength of cooperativity	Cooperativity determination data
reaction-orders.xlsx	Determination of reaction orders.	Reaction order determination data
binding_data_sq.xlsx	A tricky protein binding.	Binding data for a tricky protein.
exp_decay_data.xlsx	—	Exponential decay data for regression exercises.

3 Week 47

Dataset	Exercise	Description
deter-reacti-order-activ.csv	Determination of reaction order and activation energy	Reaction order and activation energy data
design-enzyme-kineti-exper.xlsx	Design of an enzyme kinetics experiment	Enzyme kinetics experimental design data
analys-data-set-obeyin.xlsx	Analysis of a data set obeying Michaelis-Menten kinetics	Data set analysis for enzyme kinetics
kinetics_data.xlsx	Lab Exercise Analysis: Michaelis Menten	Use your own data if you have it, use this dataset if you don't or your data is of poor quality.

4 Week 48

Dataset	Exercise	Description
enzyme-inhib-i.xlsx	Enzyme inhibitors (I)	Enzyme inhibition data (part I)
enzyme-inhib-ii.xlsx	Enzyme inhibitors (II)	Enzyme inhibition data (part II)
enzyme-behav-atcase.csv	Enzymatic behaviour of the enzyme ATCase	Aspartate transcarbamoylase enzyme behavior data

5 Week 49

Dataset	Exercise	Description
deter_delta_h_data.xlsx	Determination of ΔH° for a reaction from the van't Hoff plot.	Determination of ΔH° for a reaction dataset (Excel format)
diff_q_keq_data.xlsx	Difference between Q and K_{eq}	Dataset of time and product concentrations.

6 Additional Datasets

These datasets are available but not yet assigned to specific weeks.

Dataset	Exercise	Description
chlorophyll_adsorption.xlsx	—	Chlorophyll absorption data (Excel format)
reverse_reaction.xlsx	—	Reversible reaction kinetics data (Excel format)